# Renjith Kumar Ramachandran

Experimental Physicist | Laser-Matter Interaction | Diagnostics | Automation | Advanced Manufacturing
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## PROFESSIONAL SUMMARY

An experimental physicist with expertise in pulsed laser ablation of thin films, Plasma diagnostics, pump–probe measurements and spectroscopic analysis. I have developed an automated diagnostic setup for time-resolved studies of laser-ablation of thin films and ultrafast laser–matter interactions. I have extensive experience in vacuum-based experimental platforms, particularly laser–surface interaction, plasma devices diagnostics development and instrumentation.

## CORE TECHNICAL SKILLS

- **Laser & Diagnostics:** High-power nanosecond lasers(Powerlite 9030 continuum), pump-probe diagnostics, plasma/ surface dynamics, LIBS, ICCD imaging, streak cameras
- **Spectroscopy & Metrology:** McPherson 2061 spectrometer, Horiba iHR360, micro-Raman spectroscopy, Michelson interferometry, Talbot interferometry, high-gain detector operation (ICCD cameras, streak cameras)
- **Systems & Lab Infrastructure:** Vacuum systems, optical alignment, beam diagnostics, PLD concept and design, chamber operation, system maintenance and different probe development including Langmuir, B-dot and Emissive probes.
- **Modeling & Coding:** Python, C++, MATLAB, LAMMPS, COMSOL Multi-physics, CAD design: NX9, Solid Edge, Solid works, Shapr 3D, Optical computational tools Zemax OpticStudio
- **Engineering & Soft Skills:** Experimental troubleshooting, data analysis, interpretation, scientific writing, mentoring, presentation, cross-functional collaboration

## RESEARCH & EXPERIMENTAL EXPERIENCE

**PhD Research Scholar, Institute for Plasma Research, Gujarat, India | Jan 2022 - Present**

- Working on laser-induced surface modification and ablation with a focus on transient plasma dynamics, melt evolution and redeposition processes.
- Developed an innovative automated pump–probe diagnostic system for investigating thin-film melting dynamics and plasma expansion during laser ablation studies.
- Studied laser-driven dewetting, evaporation, and surface morphology evolution in thin-film systems under high-fluence laser irradiation.
- Used high-gain detection and spectroscopic diagnostics to study laser-produced plasmas and surface response.
- Built practical experience in vacuum operation, optical setup alignment, and troubleshooting of complex laboratory instrumentation.
- Studies nonlinear phenomena such as stimulated Brillouin scattering for pulse compression.

**M.Sc. Physics Project, University of Kerala / IIT Madras, India**

- Estimated focal length of lenses and linear thermal expansion coefficient of aluminium using interferometric techniques.
- Worked with Talbot interferometry and Michelson interferometry for precision optical measurement and analysis.

**B.Sc. Physics Project, University of Kerala, India**

- Studied phosphoric-acid-doped conducting polymer using thermogravimetric and differential scanning calorimetric analyses.
- Gained early exposure to material characterization and thermal analysis methods.

## SELECTED PUBLICATIONS

1. **R. Kumar R.** et al., “Effect of ambient on the dynamics of re-deposition in the rear laser ablation of a thin film,” Optics & Laser Technology 181 (2025) 111954 https://doi.org/10.1016/j.optlastec.2024.111954
2. **R. Kumar R.** et al., “Transient Dynamics of Front and Rear Ablation in Aluminium Thin Films,” (Under communication with journal of applied physics a. doi: https://doi.org/10.2139/ssrn.6998057)
3. **R. Kumar R.** et al., “Development of a versatile modular system to study the ablation and plume expansion of a thin film” Submitted to Review of scientific instrumentation.
4. B.R. Geethika,., **R Kumar R** et al., “Spatio-temporal dynamics of anisotropic emission from nano-second laser produced aluminium plasma,” Journal of Analytical Atomic Spectrometry 38(11) (2023) 2477-2485. DOI: 10.1039/D3JA00228D
5. B.R. Geethika,., **R Kumar R** et al., “Effect of polarization on spectroscopic characterization of laser produced aluminium plasma,” Spectrochimica Acta Part B 221 (2024) 107033. 10.1016/j.sab.2024.107033
6. B. R. Geethika,., **R Kumar R** et al. “A comprehensive study on the line profiles and Stark widths of ionic transitions from laser-produced aluminum plasma.” *J. Appl. Phys.* 21 May 2026; 139 (19): 193303. https://doi.org/10.1063/5.0324581

## SOFTWARE DEPOSITARY

1. **R. Kumar R**, J. Thomas, Automated control, acquisition and analysis software for nanosecond pump-probe laser ablation experiments (2026). doi:10.5281/zenodo. 21108254. URL https://github.com/renjithkumarcpd-dev/ Pump-probe-laser-ablation-of-thin-film

### Manuscript under Preparation

1. **R. Kumar R.** et al., — "Ostwald ripening and dewetting phenomena during laser ablation of thin films."
2. **R. Kumar R.** et al.,— "Laser surface structuring using redeposition in rear ablation of thin film"
3. **R. Kumar R.** et al.,— "Laser cleaning of optical components"

## EDUCATION

**PhD in Physics -** Institute for Plasma Research / Homi Bhabha National Institute, Mumbai | 2022 - Present
**M.Sc. in Physics -** University of Kerala | 2017 - 2019 | Aggregate: 81%
**B.Sc. in Physics -** University of Kerala | 2014 - 2017 | Aggregate: 85%

## AWARDS & ACHIEVEMENTS

- Best Oral Presentation Award for "Study of Laser Ablation Dynamics of Thin-Film Using an Innovative Pump–Probe Diagnostic Setup" in the PSC Colloquium, Guwahati, India, July 2026.
- Best poster award for "Studying the melt dynamics of a thin-film using a probe laser," International Conference on Light Matter Interaction & Ultrafast Processes, MG University, Kerala (March 2024).
- Qualified National entrance test "Joint Entrance Screening Test (JEST)" in 2021.

## SELECTED CONFERENCE PRESENTATIONS

- Oral presentation "Study of Laser Ablation Dynamics of Thin-Film Using an Innovative Pump–Probe Diagnostic Setup" in the **PSC Colloquium, Guwahati, India, July 2026.**
- Oral presentation: "Studying the dynamics of laser cleaning of optical components," **49th ITPEA Diagnostics TG Meeting, Gandhinagar (April 2026).**
- Oral presentation: "Study of Re-deposition Dynamics of Thin Film in Rear Ablation Geometry," **51st EPS Conference on Plasma Physics, Vilnius (July 2025).**
- Oral presentation: "A pump-probe study of laser ablation of a thin film in rear ablation geometry," **iCOLD25, IIT Hyderabad (March 2025).**

## TEACHING & MENTORING

- Guiding project student on "laser surface structuring"
- Guided a summer school project in 2024 on "pulse compression using Stimulated Brillouin Scattering (SBS)", which received the Best Project Award.
- Delivered talks on physics and research topics at weekly department seminar sessions.
- Assisted in demonstration of practical experiments during M.Sc. laboratory sessions.

## PROFESSIONAL PROFILES

**ResearchGate:** researchgate.net/profile/Renjith-Kumar-R
**Google Scholar:** scholar.google.com/citations?user=zZOtZ8AAAAAJ&hl=en
**LinkedIn:** linkedin.com/in/renjith-kumar-67b083212

## REFERENCES

- Dr. Jinto Thomas, Head, Laser Diagnostics Section, Institute for Plasma Research; Assistant Professor, HBNI (jinto@ipr.res.in).
- Dr. Hem Chandra Joshi, Professor (Retd.), Institute for Plasma Research (hem_sup@yahoo.co.uk).
- Dr. Ratheesh R, Assistant Professor, Department of Physics, S.N. College Punalur, Kerala University (ratheeshrphysics@gmail.com).